

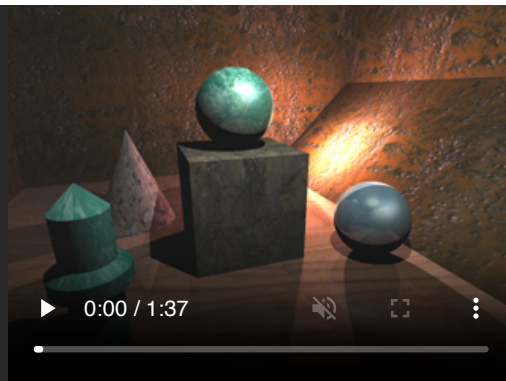
Inigo Quilez :: articles :: introduction to raytracing - 2001



How many raytracers have you written? I think I must have done 50 or so by now, some really simple and specific for a demo, others more generic and one even professional. This is just one more of those raytracers, done during 2003 after I read read about Renderman (by Pixar) and I encountered the word "shader" for the first time in my life. Well, then since everything has been about shaders around me.

This page contains some images and a video of the first raytracer I wrote that supported light and surface shaders as standalone little programs to run during surface and light evaluation. I did everything in software rendering of course, and I implemented all of the basic shaders I could find in that paper. This paper by Pixar was such a true revelation to me, and the reason for me to buy a graphics card that same week and start learning OpenGL and hardware shaders the month after (which result was [Paradise](#)). The two weeks test raytracer contained the following features:

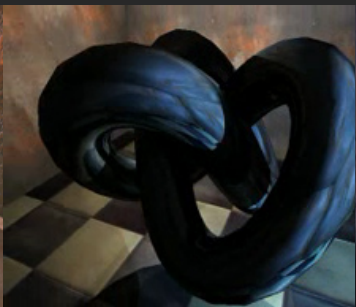
- Raster space antialiasing
- Time antialiasing
- Camera exposition simulation
- Surface shaders (procedural texturing and lighting)
- Volume shaders (procedural volumetric effects, see first video)
- Reflections, refractions, shadows
- Able of raytrace spheres, cylindres, cubes, planes, cones, revolution objects, 3d julias and polygonal meshes.
- Lame bump mapping



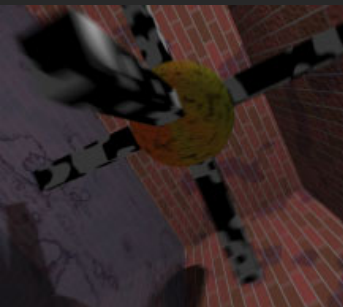
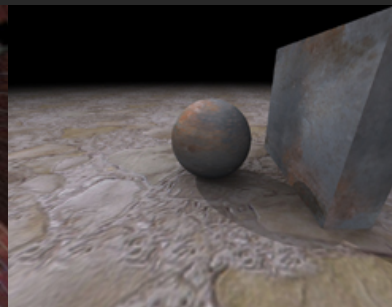
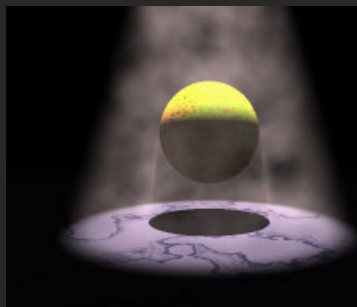
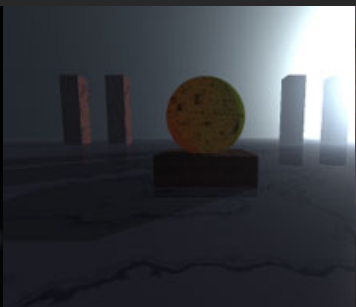
Raytracing experiments of 2001 and 2003



Reflecons - 2001



Polyon meshes - 2001

Procedural shader,
motion blur - 2003Ambient occlusion -
2003Volumetric raymarching
- 2003

Light scattering - 2003

Julia sets
(raymarching) - 2001Julia sets
(raymarching) - 2001

inigo quilez - learning computer graphics since 1994